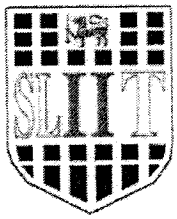
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Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Information Technology

Final Examination
Year 2, Semester 2 (2025)

IT2070 – Data Structures and Algorithms

Duration: 2 Hours

June 2025

Instructions to Candidates:

- This paper has 4 questions.
- Exam commences with 10 minutes reading period.
- Answer all questions in the booklet given.
- The total marks for the paper is 100.
- This paper contains 7 pages, including the cover page.
- Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.

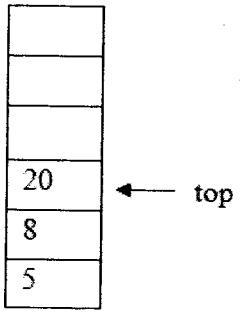
Question 1 **25 marks**

(1) "A stack can be created using a linked list". Do you agree with this statement? Briefly discuss your opinion. (3 marks)

(2) "The performance of a circular queue is greater than a linear queue". Briefly explain your opinion about this statement. (2 marks)

(3) How does a queue can be used to store the keystroke data as a user type at the Keyboard? (2 marks)

(4) Consider the following stack data structure. Draw the stack frames after executing the following operations and indicate the values of top and count. (6 marks)



(a) push(50)
(b) pop()
(c) peek()
(d) pop()
(e) pop()
(f) pop()

(5) If the same operations have been done for a linear queue with the size of 6 with the same content, state the values of front, rear and count for each operation. (3 marks)

Note that the push(), pop() and peek() operations of a stack can be aligned with insert(), remove() and peekFront() methods in a queue. Consider front points to element 5 and rear points to element 20.

(6) Write a main program in Java to remove all the occurrences of a given element of a stack. (9 marks)

Note that the size of the stack, content and the element to be removed should be taken as keyboard inputs.

Assume that the StackX class has been already implemented with push(), poop(), peek(), isEmpty() and isFull() methods.

Input :

Stack elements:

10 4 10 5 10

Element to be removed:

10

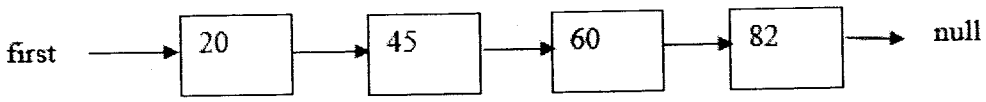
Output:

4 5

Question 2 **25 marks**

(1) Compare and contrast array and a linked list. (2 marks)

(2) Consider the linked list given below and state the output generated after executing each of the code segments. (6 marks)



(a)

```
Link current = first;
while (current != null)
{
        System.out.print (current.iData + " ");
        current = current. next;
}
```

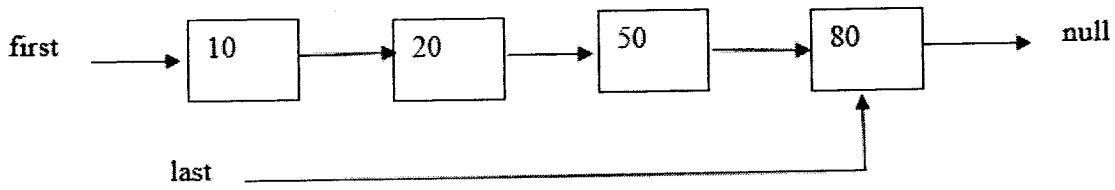
(b)

```
Link current = first;
current = current. next;
while (current != null)
{
        System.out.print (current.iData + " ");
        current = current. next;
}
```

(c)

```
Link current = first;
while (current.next != null)
{
        System.out.print (current.iData + " ");
        current = current. next;
}
```

(3) Consider the following double-ended list. Write down the code segments for the following instances. (8 marks)



(a) Delete link 50
(b) Delete and return 10
(c) Insert 30 after 10

(4) Compute the number of nodes of full binary tree with height 3. (1 mark)

(5) Consider the following set of nodes.

210,56,120,320,450,150,256,510,289,420

(a) Arrange these nodes into a binary search tree and draw that tree. (5 marks)

(b) Indicate the depth levels in the tree that you have drawn in (5) (a). (2 marks)

(c) What is the height of this tree? (1 mark)

Question 3 **25 marks**

(1) State whether the following statements are either TRUE OR FALSE. (5 marks)

(a) Asymptotic notations are the most practical way of representing the complexity of an algorithm.
(b) Complexities of Heapsort and Mergesort algorithms are same.
(c) Complexities of Insert() and Extract() of a priority queue are same.
(d) Divide and Conquer Method is an algorithm designing technique.
(e) Mergesort is a not recursive algorithm.

(2) Express the T(n) of following pseudocodes using RAM model. (6 marks)

(a)

```
a = 5
while a > 0
        a = a - 2
        print a
```

(b)

```
for i = 1 to 5
        for j = 1 to n-1
                a = A[i] + i * j
                print a
```

(3) "The Complexities of Quick Sort best case and Merge Sort are equal". Do you agree with this statement? Briefly explain the reason. (2 marks)

(4) What is the recurrence equation for the Merge Sort algorithm? (2 marks)

(5) Solve the above (4) recurrence equation using Master Theorem. (3 marks)

$$T(n) = \begin{cases} \Theta\left(n^{\log_b a}\right) & f(n) = O\left(n^{\log_b a - \varepsilon}\right) \rightarrow f(n) < n^{\log_b a} \\ \Theta\left(n^{\log_b a} \lg n\right) & f(n) = \Theta\left(n^{\log_b a}\right) \rightarrow f(n) = n^{\log_b a} \\ \Theta\left(f(n)\right) & f(n) = \Omega\left(n^{\log_b a + \varepsilon}\right) \rightarrow f(n) > n^{\log_b a} \\ & \text{if } af(n/b) \le cf(n) \text{ for } c < 1 \text{ and large } n \end{cases}$$

(6) Illustrate the operations of Heap_Insert(A, 275) for heap A with given set of elements. (For illustration process assign the values only once and then use a diagrammatic approach to reach the answer.) (7 marks)

A

300	189	250	170	110	230	245

```
HEAP_INSERT(A, key)
1 A.heap_size = A.heap_size + 1
2 i = A.heap_size
3 while i > 1 and A[PARENT(i)] < key
4     A[i] = A[PARENT(i)]
5     i = PARENT(i)
6 A[i] = key
```

Question 4 **25 marks**

(1) Using suitable example, derive the time complexity of the best case of Naïve String Matching algorithm. (4 marks)

(2) Compute the number of valid and invalid shifts for the following Text and the Pattern with the use of Naïve String Matching algorithm. (4 marks)

T = 1095680760
P = 50

(3) Derive the number of valid and spurious hits if you apply the same Text and the Pattern in (2) with Rabin Carp algorithm with the modulo value (q) = 10. (5 marks)

(4) Which computation is better comparing to Naïve String Matching and Rabin Carp algorithms? Briefly explain your answer. (3 marks)

(5) Draw the state transition diagram for Pattern P = *ccb* along with the input alphabet = {b,c}. (9 marks)

---------- ***End of the Paper***----------